



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

AUGS/ AGSR Division

I SEMESTER 2026-27 COURSE HANDOUT

Date: 01.08.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course. **Please read EVERYTHING thoroughly.**

Course No.	EEE / ECE / INSTR / CS F342	
Course Title	Digital Design	
Instructor (lec)	Saurabh Gandhi (IC) [https://tinyurl.com/f215-contact ; office 2220-M] G. S. S. Chalapathi Ramesh Sambangi	
Instructors (tutorial)	Saurabh Gandhi Arnab Hazra Puneet Mishra Ramesh Sambangi Pranjal Vyas	Nitin Chaturvedi Balamurugan P (RS) Gopal Singh Phartiyal Tejasvi Alladi
Instructors (lab)	Shreya Chandravanshi (RS) Rejaul (RS) Antony David (RS) Devanshu Sahoo (RS) Suvrojoyti Biswas (RS) Balamurugan P. (RS) Aniket Kumar (RS) Aishwarya B. Mudennavar (RS) Nimish Kumar (RS) Deepak Pareek (RS) Rohit Kumar (RS)	Raja Piyush (RS) Nilesh Jaiswal (RS) Harshvardhan Singh (RS) Harshita Goyal (RS) Keerthana P. Balakrishnan (RS) Harshita Saroj (RS) Ankit Sharma (RS) Parmod (RS) Vipin Kumar Saraogi (RS) Raman (RS)

1. Course description

Boolean Algebra & logic minimization; combinational logic circuits: arithmetic circuit design, Design using MSI components; Sequential Logic Circuits: flip flops & latches, registers and counters, Finite state machine; HDL Implementation of Digital circuits; Digital Integrated Circuits; Programmable logic devices; Memory organization; Algorithmic State machine; Introduction to computer organization; The course will also have laboratory component on digital design.

2. Textbook(s)

T1: M. Moris Mano and Michael D. Ciletti "Digital Design," PHI, 6th Edition, 2018.

T2: G. Raghurama, S. Gurunarayanan, S. Mohan, Karthik, Pawan K. Ajmera, "Laboratory Manual", 2022

3. Reference book(s) and online resources

R1: Donald D. Givonne, "Digital Principles and Design" TMH, 2003

R2: W. Stallings, "Computer Organization and Architecture" PHE, 10th Edition, 2015



4. Course Objectives

1.	Understand	• Explain binary/BCD number representations and perform base conversions • State the postulates of Boolean algebra and the meaning of minterms, maxterms, SOP/POS forms • Describe the operation of standard combinational building blocks (adders, decoders, multiplexers, comparators, ALUs) and standard sequential elements (latches, flip-flops, registers, counters) • Describe the characteristics of digital logic families (BJT, TTL, MOS) and PLD architectures • Explain algorithms for signed multiplication (Booth's) and division
2.	Apply	• Simplify Boolean functions using K-maps (including don't-cares, PI/EPI) and the Quine-McCluskey method • Analyze and derive state tables/diagrams for a given sequential circuit • Compare design tradeoffs between ripple-carry vs. carry-lookahead adders, and between different logic family implementations • Trace/debug simple Verilog modules against their intended combinational or sequential behavior
3.	Create	• Design combinational circuits (use of adders, decoders, mux/demux, PLDs) to meet given specifications • Design sequential circuits (use of counters, registers, shift registers) from a given behavioral description • Design a digital system from an algorithmic state machine (ASM) chart • Implement and verify combinational/sequential designs in Verilog

5. Module Objectives

	Name	Objectives
1.	Working in Binary	• Represent and convert numbers across binary/BCD systems, and apply Boolean algebra postulates to manipulate logic expressions • Simplify Boolean functions using K-maps (minterms, maxterms, don't-cares, PI/EPI) into minimal SOP/POS forms • Apply the Quine-McCluskey method as a tabular alternative to K-map simplification
2.	Combinational Building Blocks	• Explain the operation of standard combinational blocks: adders/subtractors, decoders, multiplexers, comparators, ALUs, PLDs • Design combinational circuits from specifications using these



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

AUGS/ AGSR Division

		building blocks (including NAND/NOR-only implementations)
3.	Sequential Design and Building Blocks	- Analyze latches, flip-flops, and sequential circuits via state tables/diagrams - Design sequential circuits — counters, registers, shift registers — from a given behavioral specification
4.	HDL (Verilog)	Implement and verify combinational and sequential circuit designs in Verilog
5.	Algorithmic State Machines	Construct ASM charts for a given system behavior and translate them into a digital circuit design
6.	Digital Logic Families	- Describe the electrical and switching characteristics of BJT, TTL, and MOS-based logic families - Compare logic families on performance tradeoffs (speed, power, noise margin)
7.	Computer Arithmetic	Explain and trace algorithms for signed multiplication (Booth's) and division

6. Lecture Schedule

See detailed course schedule (lectures, tutorials, labs and exams) [here](#). FAQs [here](#).

7. Evaluation components

Component	Duration	Total Marks	Nature of component	Date and Time
Midsemester exam	90 mins	75	CB/OB	See tentative schedule above
Comprehensive exam	180 mins	105	CB/OB	
Quizzes (best 2 / 3)	45 mins each	30	CB	
Lab exam ^{\$}	N/A	50	OB	
Class participation [%]	N/A	20 + 5	OB	All classes
Lab performance [#]	N/A	15	OB	All labs

[#]1.5 marks each, best 10 out of 12 (for attendance and completion as defined for each lab)

[%]based on the engagement score described below

^{\$}[IPC lab] Converting a circuit to Verilog (15 marks), writing an appropriate test bench (15 marks) + [Digital lab] Clean and functional physical implementation (20 marks)



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani

Pilani Campus

AUGS/ AGSR Division

8. Chamber consultation hour

To be announced in class.

9. Make-up policy

Makeup will be granted to extremely genuine cases only (like hospitalization, etc.), provided the IC has been informed.

10. Minimum performance requirement for valid grade

A student should obtain 25% of the average of the class, or 10% of the course total marks, whichever is lower, to clear the course. If any student gets marks lower than the prescribed standard mentioned above, he/she may be awarded an NC report.

11. Course logistics

- a. **Requests and student communications:** All requests and administrative queries (logistics, make-ups, absence permissions, doubts, meeting time, complaints etc.) should be communicated through this Google Form **ONLY**: <https://forms.gle/B7XuYA6gA28M8Pg26>. **ANY** communication sent over email **WILL NOT** be acted upon.
- b. **CMS:** We will be using Google Classroom for content management. **Please join the classroom using this link:** <https://classroom.google.com/c/ODcxMTQyMzA1NjYy?cjc=bwosvprd>
Note that we are not using Nalanda. All announcements and material will be shared via Google Classroom only.
- c. **Interaction App:** We will use a web app for interacting in the lectures to answer questions in the class. Your responses will be used for recording attendance and the “class participation” score included as an evaluated component.
The app shows three scores, which are used as follows:
Engagement score: mostly based on attempting questions, although it will check *some* questions for correctness also to ensure you are not answering randomly.
Basic competence: based on correctness of answers, weighted heavily on easy questions. *Not included in evaluation.*
Mastery: based on correctness of answers, weighted heavily by advanced questions. *Not included in evaluation.*
Access and registration: <https://student-ces.vercel.app/>.

Instructor-in-charge
CS F215 / ECE F215 / EEE F215 / INSTR F215